

Spectral Obstructions from Structural Stability

A substantial partial result for arXiv:1902.04386

Automated mathematics run `fa_banach_001`, lane 8

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Status. Candidate substantial partial result, likely valid, pending human review. The ordinary structural-stability questions remain open.

1 Source questions and scope

Bernardes Jr. and Messaoudi ask whether shadowing and structural stability imply one another (Question 13), whether an expansive structurally stable invertible operator must be hyperbolic (Question 15), and whether hyperbolicity coincides with structural stability relative to $GL(X)$ (Question 16) [1]. The first source crop records Question 13; the second records Questions 15 and 16.

question.

Question 13. Is it true that every invertible operator on a Banach space that has the shadowing property is structurally stable? How about the converse?

Figure 1: Question 13 in [1, p. 12].

The previous results motivate the following question.

Question 15. If an invertible operator on a Banach space is expansive and structurally stable, is it necessarily hyperbolic?

The following question also seems to be open.

Question 16. Do hyperbolicity and structural stability relative to $GL(X)$ coincide?

Figure 2: Questions 15 and 16 in [1, p. 13].

The present note proves two general spectral obstructions. The first uses ordinary structural stability and removes the spectral point 1. The second uses the near-identity conjugacy in *strong* structural stability and removes every unimodular eigenvalue of the adjoint. This gives a complete positive answer to Question 15 with “structurally stable” replaced by “strongly structurally stable”, on arbitrary complex Banach spaces. For ordinary stability, it identifies the only residual-spectrum regime in which a counterexample can remain.

2 Definitions

Let X be a real or complex Banach space and $T \in GL(X)$. Following [1], T is *structurally stable* if some $\varepsilon > 0$ has the following property: whenever $\alpha \in C_b(X; X)$ is bounded and Lipschitz with

$$\|\alpha\|_\infty \leq \varepsilon, \quad \text{Lip}(\alpha) \leq \varepsilon,$$

the homeomorphisms T and $T + \alpha$ are topologically conjugate. It is *strongly structurally stable* if, for every $\gamma > 0$, the tolerance can be chosen so that a conjugacy h can additionally satisfy $\|h - I\|_\infty \leq \gamma$.

For a complex Banach space, $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, $\sigma_p(T^*)$ is the point spectrum of the adjoint, and $\sigma_a(T)$ is the approximate point spectrum. An invertible T is hyperbolic when $\sigma(T) \cap \mathbb{T} = \emptyset$.

3 Proof intuition

Ordinary stability already controls constant affine perturbations. Since T fixes the origin, every conjugate map $T + a$ must have a fixed point; this says exactly that every small a belongs to $\text{Ran}(I - T)$.

Strong stability supplies much more: a conjugacy uniformly close to the identity. If $T^*f = \lambda f$ with $|\lambda| = 1$, follow under that conjugacy an orbit whose f -coordinate is the circle $r\lambda^n$. A bounded Lipschitz perturbation, supported in a thin annulus of the scalar f -coordinate, changes the quotient multiplier from λ to $(1 + c)\lambda$. Closeness of the conjugacy traps the quotient orbit in the annulus, while the new multiplier forces exponential radial growth. These two facts are incompatible.

For an expansive ordinarily stable operator, the source paper proves uniform expansivity [1, Theorem 6(a)]. Thus a nonhyperbolic unit-circle spectral point cannot be approximate spectrum: it has closed proper range and becomes an adjoint eigenvalue. The ordinary result removes $\lambda = 1$; the strong result removes all such λ .

4 Main results

Theorem 1 (Ordinary stability forces affine solvability). *Let $T \in GL(X)$ be structurally stable. Then $I - T$ is surjective. If, in addition, T is expansive, then $I - T$ is invertible; equivalently, $1 \in \rho(T)$.*

Proof. Choose a structural-stability tolerance $\varepsilon > 0$. For any $a \in X$ with $\|a\| \leq \varepsilon$, the constant map $\alpha_a(x) = a$ satisfies $\|\alpha_a\|_\infty \leq \varepsilon$ and $\text{Lip}(\alpha_a) = 0$. Hence there is a homeomorphism $h : X \rightarrow X$ such that

$$h \circ T = (T + \alpha_a) \circ h.$$

Evaluating at 0 gives

$$h(0) = Th(0) + a, \quad \text{so} \quad (I - T)h(0) = a.$$

Thus $\text{Ran}(I - T)$ contains the ball εB_X . Since it is a linear subspace, $\text{Ran}(I - T) = X$.

If T is expansive and $(I - T)x = 0$, then $T^n x = x$ for every $n \in \mathbb{Z}$. After normalizing a nonzero such x , this contradicts expansivity. Hence $I - T$ is injective. The bounded inverse theorem now gives $(I - T)^{-1} \in \mathcal{B}(X)$. $\square$

Theorem 2 (Strong stability excludes unimodular adjoint eigenvalues). *Let X be a complex Banach space and let $T \in GL(X)$ be strongly structurally stable. Then*

$$\sigma_p(T^*) \cap \mathbb{T} = \emptyset.$$

Proof. Suppose instead that $T^*f = \lambda f$ for some $f \in X^* \setminus \{0\}$ and $\lambda \in \mathbb{T}$. Normalize $\|f\| = 1$ and choose $v \in X$ with $f(v) = 1$. Fix $\gamma > 0$ and put $r = 4\gamma$. Let $\rho : [0, \infty) \rightarrow [0, 1]$ be a Lipschitz function such that

$$\rho(t) = 1 \quad (t \in [r - \gamma, r + \gamma]), \quad \rho(t) = 0 \quad (t \notin [r - 2\gamma, r + 2\gamma]).$$

For $c > 0$, define

$$\alpha_c(y) = c\lambda f(y)\rho(|f(y)|)v.$$

The scalar map $z \mapsto z\rho(|z|)$ is bounded and Lipschitz. Consequently $\|\alpha_c\|_\infty$ and $\text{Lip}(\alpha_c)$ tend to zero with c . Choose $c > 0$ small enough that α_c is admitted by strong structural stability (and, if necessary, small enough that $T + \alpha_c$ is a homeomorphism). We obtain a homeomorphism $h = I + u$ satisfying

$$h \circ T = (T + \alpha_c) \circ h, \quad \|u\|_\infty \leq \gamma.$$

Take $x = rv$ and set $q_n = f(h(T^n x))$ for $n \in \mathbb{Z}$. Since $f(T^n x) = \lambda^n f(x) = \lambda^n r$,

$$|q_n - \lambda^n r| = |f(u(T^n x))| \leq \gamma.$$

It follows that $|q_n| \in [r - \gamma, r + \gamma]$ for every n , so the cutoff is identically one along this orbit. Applying f to the conjugacy equation yields

$$\begin{aligned} q_{n+1} &= f(Th(T^n x)) + f(\alpha_c(h(T^n x))) \\ &= \lambda q_n + c\lambda q_n = (1 + c)\lambda q_n. \end{aligned}$$

Therefore $|q_n| = (1 + c)^n |q_0|$ for $n \geq 0$. But $|q_0| \geq r - \gamma > 0$, whereas the annulus estimate gives $|q_n| \leq r + \gamma$ for every n . This is a contradiction. $\square$

Corollary 3 (Sharp expansive consequences). *Let X be a complex Banach space.*

- (a) *If $T \in GL(X)$ is expansive and structurally stable, then $1 \in \rho(T)$. If T is not hyperbolic, every $\lambda \in \sigma(T) \cap \mathbb{T}$ satisfies $\lambda \neq 1$, $T - \lambda I$ is bounded below with closed proper range, and $\lambda \in \sigma_p(T^*)$.*
- (b) *If $T \in GL(X)$ is expansive and strongly structurally stable, then T is hyperbolic.*
- (c) *If T is expansive and ζT is structurally stable for every $\zeta \in \mathbb{T}$, then T is hyperbolic.*

Proof. The first statement in (a) is Theorem 1. Theorem 6(a) of [1] says that an expansive structurally stable operator is uniformly expansive. Hence $\sigma_a(T) \cap \mathbb{T} = \emptyset$. If $\lambda \in \sigma(T) \cap \mathbb{T}$, the operator $T - \lambda I$ is bounded below, so its range is closed. It is not surjective, since $\lambda \in \sigma(T)$, and therefore Hahn–Banach gives a nonzero $f \in X^*$ annihilating its range. Thus $T^*f = \lambda f$. The fact that $\lambda \neq 1$ follows from Theorem 1.

Part (b) follows because Theorem 2 rules out the adjoint eigenvalue forced by every putative unit-circle spectral point in (a).

For (c), ζT is expansive whenever T is. Applying Theorem 1 to ζT shows that $I - \zeta T$ is invertible for every $\zeta \in \mathbb{T}$. Equivalently, $\sigma(T) \cap \mathbb{T} = \emptyset$. $\square$

5 What remains open

Theorem 1 does not answer Question 13 or the ordinary version of Question 15. An arbitrary conjugacy supplied by ordinary structural stability need not be close to the identity, boundedly displaced, homogeneous, or equivariant under scalar rotations. The annular resonance in Theorem 2 uses the estimate $\|h - I\|_\infty \leq \gamma$ at exactly one indispensable point.

Eight focused upgrade routes were checked: affine perturbations; transfer to ζT ; rescaling an arbitrary conjugacy; periodic/root-of-unity perturbations; difference-operator selection for shadowing; an upgrade through topological stability; finite interpolation by small Lipschitz homeomorphisms; and linear rank-one perturbations for the relative $GL(X)$ question. In each failed route, the same obstruction remains: a uniformly expansive nonhyperbolic operator can

have closed proper range for $T - \lambda I$ and an exact eigenfunctional of T^* , but no approximate eigenvectors of T from which to build an arbitrarily small-Lipschitz resonant orbit.

Thus a counterexample to ordinary Question 15, if one exists, must live entirely in the nontrivial residual-spectrum regime described in Corollary 3(a). No new conclusion about Question 16 is claimed.

6 Verification and bounded novelty search

The proof is symbolic. The main verification points are:

- constant perturbations are bounded with Lipschitz constant zero, and conjugacy sends the fixed point 0 of T to a fixed point of $T + a$;
- $z \mapsto z\rho(|z|)$ is globally bounded and Lipschitz because ρ has compact support;
- the conjugacy orientation gives $q_{n+1} = (1 + c)\lambda q_n$;
- uniform expansivity removes the approximate point spectrum from $\mathbb{T}$, so a remaining spectral point has closed proper range and produces an adjoint eigenvector.

The run’s registry, solution, attempt, and proof-gap indexes were searched by arXiv id and by the core phrases. A bounded web/arXiv search through 17 August 2026 checked the source paper, Bayart’s weighted-shift characterization [2], the composition-operator work [3], the 2024 shadowing paper [4], and the 2025 topological-stability paper [5]. The latter proves that topological stability implies shadowing on Banach spaces, but its near-identity semiconjugacy hypothesis is stronger than the ordinary structural stability used here. No exact statement of Theorems 1–2 or the general strong-stability extension in Corollary 3(b) was found. This is not a proof of novelty.

Human-review recommendation. Promote for expert review as a substantial partial result. The most important check is the annulus recurrence in Theorem 2; the second is whether the direct strong-stability corollary has appeared under different terminology.

References

- [1] N. C. Bernardes Jr. and A. Messaoudi, *Shadowing and structural stability in linear dynamical systems*, arXiv:1902.04386 (2019); published as *Shadowing and structural stability for operators*, Ergodic Theory Dynam. Systems 41 (2021), 961–980.
- [2] F. Bayart, *Two problems on weighted shifts in linear dynamics*, arXiv:2101.02989 (2021), Proc. Amer. Math. Soc. 149 (2021), 5255–5270.
- [3] M. Maiuriello, *Expansivity and strong structural stability for composition operators on L^p spaces*, arXiv:2206.00353 (2022), Banach J. Math. Anal. 16 (2022).
- [4] N. C. Bernardes Jr. and A. Peris, *On shadowing and chain recurrence in linear dynamics*, Adv. Math. 441 (2024), 109539.
- [5] N. C. Bernardes Jr., B. M. Caraballo, U. B. Darji, V. V. Fávaro, and A. Peris, *Generalized hyperbolicity, stability and expansivity for operators on locally convex spaces*, arXiv:2403.02843, J. Funct. Anal. 288 (2025), 110696.